

Python Assignment - 3

School of Engineering & Technology

Department: SOET	Session: 2025-26
Program: MCA (AI & ML)	Semester: 1
Course Code: ETCCPP171	College Rollno: 2501940002
Course Name: Programming for Problem Solving Using Python	
Submitted by: Ram Kumar Singh	Faculty: Ms. Neha Kaushik

GitHub Link : https://github.com/RamOkr0singh/Library_Managment_system_0002.git

Assignment: Student Attendance Tracker

INPUT :-

```
# -----
# Name: Ram Kumar Singh
# Date: 25 Nov 2025
# Assignment 03 - Library Management System
# -----

import os
import json

# =====
#                      BOOK ENTITY
# =====

class Book:
    def __init__(self, book_id, title, author, available=True):
        self.book_id = book_id
        self.title = title
        self.author = author
        self.available = available

    def to_dict(self):
        return {
            "book_id": self.book_id,
            "title": self.title,
            "author": self.author,
            "available": self.available
        }

# =====
#                      MEMBER ENTITY
# =====

class Member:
    def __init__(self, member_id, name, borrowed_books=None):
        self.member_id = member_id
        self.name = name
```

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        self.borrowed_books = borrowed_books if borrowed_books else []

    def to_dict(self):
        return {
            "member_id": self.member_id,
            "name": self.name,
            "borrowed_books": self.borrowed_books
        }

# =====
#                               LIBRARY SYSTEM
# =====
class Library:
    def __init__(self, books_file="books.json",
members_file="members.json"):
        self.books_file = books_file
        self.members_file = members_file
        self.books = self._load_file(self.books_file)
        self.members = self._load_file(self.members_file)

# -----
# File Utilities
# -----
    def _load_file(self, file_path):
        if not os.path.exists(file_path):
            return {}
        with open(file_path, "r") as f:
            return json.load(f)

    def _save_file(self, file_path, data):
        with open(file_path, "w") as f:
            json.dump(data, f, indent=4)

# -----
# Core Library Functionalities
# -----
    def add_book(self, book):
        self.books[book.book_id] = book.to_dict()
        self._save_file(self.books_file, self.books)
        print("Book added successfully.")

    def add_member(self, member):
        self.members[member.member_id] = member.to_dict()
        self._save_file(self.members_file, self.members)
        print("Member added successfully.")

    def borrow_book(self, member_id, book_id):
        if member_id not in self.members:
            print("Invalid Member ID.")
            return

        if book_id not in self.books:
            print("Invalid Book ID.")
            return

        if not self.books[book_id]["available"]:
```

```

        print("Book is already borrowed.")
        return

    self.books[book_id]["available"] = False
    self.members[member_id]["borrowed_books"].append(book_id)

    self._save_file(self.books_file, self.books)
    self._save_file(self.members_file, self.members)

    print("Book borrowed successfully.")

def return_book(self, member_id, book_id):
    if member_id not in self.members:
        print("Invalid Member ID.")
        return

    if book_id not in self.members[member_id]["borrowed_books"]:
        print("This member did not borrow this book.")
        return

    self.members[member_id]["borrowed_books"].remove(book_id)
    self.books[book_id]["available"] = True

    self._save_file(self.books_file, self.books)
    self._save_file(self.members_file, self.members)

    print("Book returned successfully.")

# -----
# Display Functions
# -----
def show_books(self):
    print("\n--- Library Books ---")
    for book in self.books.values():
        status = "Available" if book["available"] else "Borrowed"
        print(f'{book["book_id"]}: {book["title"]} by {book["author"]} ({status})')

def show_members(self):
    print("\n--- Library Members ---")
    for member in self.members.values():
        print(f'{member["member_id"]}: {member["name"]} | Borrowed: {member["borrowed_books"]}')

# =====
#                               MAIN MENU
# =====
def main():
    library = Library()

    menu = """
===== Library Menu =====
1. Add Book
2. Add Member
3. Borrow Book
4. Return Book

```

```

5. Show All Books
6. Show All Members
7. Exit
"""
while True:
    print(menu)
    choice = input("Enter your choice: ").strip()

    if choice == "1":
        book = Book(
            input("Book ID: "),
            input("Title: "),
            input("Author: ")
        )
        library.add_book(book)

    elif choice == "2":
        member = Member(
            input("Member ID: "),
            input("Name: ")
        )
        library.add_member(member)

    elif choice == "3":
        library.borrow_book(
            input("Member ID: "),
            input("Book ID: ")
        )

    elif choice == "4":
        library.return_book(
            input("Member ID: "),
            input("Book ID: ")
        )

    elif choice == "5":
        library.show_books()

    elif choice == "6":
        library.show_members()

    elif choice == "7":
        print("Exiting... Goodbye!")
        break

    else:
        print("Invalid choice. Please try again.")

if __name__ == "__main__":
    main()

```

OUTPUT :-

```
PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL
PS R:\Coding\1. KRMU\Python\Sem 1\Library_Managment_system_0002> python -u "r:\Coding\1. KRMU\Python\Sem 1\Library_Managment_system_0002\main.py"

===== Library Menu =====
1. Add Book
2. Add Member
3. Borrow Book
4. Return Book
5. Show All Books
6. Show All Members
7. Exit

Enter your choice: 1
Book ID: 0001
Title: Usogui
Author: Toshio Sako
Book added successfully.

===== Library Menu =====
1. Add Book
2. Add Member
3. Borrow Book
4. Return Book
5. Show All Books
6. Show All Members
7. Exit

Enter your choice: 1
Book ID: 0002
Title: Vagabond
Author: Takehiko Inoue
Book added successfully.

===== Library Menu =====
1. Add Book
2. Add Member
3. Borrow Book
4. Return Book
5. Show All Books
6. Show All Members
7. Exit
```

```
EXPLORER main.py U {} books.json U X README.md
LIBRARY_MANAGMENT_SYSTEM_0002
  ~$thon Assignment 3.docx U
  {} books.json U
  main.py U
  {} members.json U
  Python Assignment 3.docx U
  Python Assignment 3.pdf U
  README.md
{} books.json > ...
1 {
2     "0001": {
3         "book_id": "0001",
4         "title": "Usogui",
5         "author": "Toshio Sako",
6         "available": false
7     },
8     "0002": {
9         "book_id": "0002",
10        "title": "Vagabond",
11        "author": "Takehiko Inoue",
12        "available": true
13    }
14 }
```

```
PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL
PS R:\Coding\1. KRMU\Python\Sem 1\Library_Managment_system_0002> python -u "r:\Coding\1. KRMU\Python\Sem 1\Library_Managment_system_0002\main.py"
Book added successfully.

===== Library Menu =====
1. Add Book
2. Add Member
3. Borrow Book
4. Return Book
5. Show All Books
6. Show All Members
7. Exit

Enter your choice: 2
Member ID: 1001
Name: Ram Kumar Singh
Member added successfully.

===== Library Menu =====
1. Add Book
2. Add Member
3. Borrow Book
4. Return Book
5. Show All Books
6. Show All Members
7. Exit

Enter your choice: 2
Member ID: Harsh Singh
Name: Harsh Singh
Member added successfully.

===== Library Menu =====
1. Add Book
2. Add Member
3. Borrow Book
4. Return Book
5. Show All Books
6. Show All Members
7. Exit

Enter your choice: 
```

```
EXPLORER
LIBRARY_MANAGMENT_SYSTEM_0002
  ~$thon Assignment 3.docx
  {} books.json
  {} main.py
  {} members.json
  {} Python Assignment 3.docx
  {} README.md

main.py U
members.json U X
README.md

{} members.json > {} 1001 > abc name
1  {
2      "1001": {
3          "member_id": "1001",
4          "name": "Ram Kumar Singh",
5          "borrowed_books": [
6              "0001"
7          ]
8      },
9      "Harsh Singh": {
10         "member_id": "Harsh Singh",
11         "name": "Harsh Singh",
12         "borrowed_books": []
13     }
14 }
```

```
PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL
PS R:\Coding\1. KRMU\Python\Sem 1\Library_Managment_system_0002> python -u "r:\Coding\1. KRMU\Python\Sem 1\Library_Managment_system_0002\main.py"

===== Library Menu =====
1. Add Book
2. Add Member
3. Borrow Book
4. Return Book
5. Show All Books
6. Show All Members
7. Exit

Enter your choice: 3
Member ID: 1001
Book ID: 0001
Book borrowed successfully.

===== Library Menu =====
1. Add Book
2. Add Member
3. Borrow Book
4. Return Book
5. Show All Books
6. Show All Members
7. Exit

Enter your choice: |
```

```
PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL
PS R:\Coding\1. KRMU\Python\Sem 1\Library_Managment_system_0002> python -u "r:\Coding\1. KRMU\Python\Sem 1\Library_Managment_system_0002\main.py"

===== Library Menu =====
1. Add Book
2. Add Member
3. Borrow Book
4. Return Book
5. Show All Books
6. Show All Members
7. Exit

Enter your choice: 5

--- Library Books ---
0001: Usogui by Toshio Sako (Borrowed)
0002: Vagabond by Takehiko Inoue (Available)

===== Library Menu =====
1. Add Book
2. Add Member
3. Borrow Book
4. Return Book
5. Show All Books
6. Show All Members
7. Exit

Enter your choice: 6

--- Library Members ---
1001: Ram Kumar Singh | Borrowed: ['0001']
Harsh Singh: Harsh Singh | Borrowed: []

===== Library Menu =====
1. Add Book
2. Add Member
3. Borrow Book
4. Return Book
5. Show All Books
6. Show All Members
7. Exit
```